

Requirements-Design Traceability Table

FR ID	Functional Requirement	Design Mapping
FR-01.1	Allow authorized users to log in using username and password	Classes: UserManagementService; Components: User Authentication
FR-01.2	Validate entered credentials against stored user records	Classes: UserManagementService; Components: User Authentication
FR-01.3	Grant access based on authenticated user's role	Classes: UserManagementService, Admin, Waiter; Components: User Authentication
FR-01.4	Deny access and display error message if authentication fails	Classes: UserManagementService; Components: User Authentication
FR-01.5	Provide "Forgot Password" feature for password recovery	Classes: UserManagementService; Components: User Authentication
FR-02.1	Allow waiter to select table number for new order	DFD Level 2.1: Process 2.1 "Select Table & Initialize Order"; Sequence Diagram: Place Order; Activity Diagram: Order Workflow; Classes: OrderManagementService, Table, Waiter; Database: D5 Tables; Components: Order Placement
FR-02.2	Display available menu items after table selection	DFD Level 2.1: Process 2.1 accessing D1 Menu Items; Sequence Diagram: Place Order – displayMenuItems(); Classes: MenuManagementService, MenuItem; Database: D1 Menu Items; Components: Order Placement

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FR-02.3	Allow waiter to add one or more menu items to order	DFD Level 2.1: Process 2.2 “Add/Update/Remove Menu Items”; Sequence Diagram: Place Order – loop for item selection; Classes: Order, OrderItem, MenuItem; Components: Order Placement
FR-02.4	Allow waiter to update or remove selected menu items before submission	DFD Level 2.1: Process 2.2 “Add/Update/Remove Menu Items”; Sequence Diagram: Modify or Cancel Order with modifications; Activity Diagram: Order Modification and Cancellation Workflow; Classes: Order, OrderItem, OrderManagementService; Components: Order Placement
FR-02.5	Allow customers to modify or cancel orders through waiter before submission	DFD Level 2.2: Process 2.9 “Process Order Cancellation”; Sequence Diagram: Modify or Cancel Order; Activity Diagram: Order Modification and Cancellation Workflow; Classes: OrderManagementService, Order; Components: Order Processing
FR-02.6	Automatically calculate 0.5% discount per 1,000 PKR when total $\geq 10,000$ PKR	DFD Level 2.1: Process 2.3 “Calculate Discount”; Sequence Diagram: Place Order – calculateDiscount() with note “Discount = 0.5% per 1,000 PKR”; Activity Diagram: Order Workflow – discount calculation; Classes: OrderManagementService; Components: Order Placement

FR ID	Functional Requirement	Design Mapping
FR-02.7	Automatically add complementary items when total $\geq$ 10,000 PKR (up to 2% value)	DFD Level 2.1: Process 2.4 “Select & Add Complementary Items”; Sequence Diagram: Place Order – selectComplementaryItems() with note “System automatically selects complementary items up to 2% of subtotal”; Activity Diagram: Order Workflow – complementary item selection; Classes: OrderManagementService, ComplementaryItem; Database: D3 Complementary Items; Components: Order Placement
FR-02.8	Select complementary items from predefined list (drinks, desserts, appetizers)	DFD Level 2.1: Access to D3: Complementary Items; Sequence Diagram: Place Order - getActiveComplementaryItems(); Classes: ComplementaryItem, MenuManagementService; Database: D3: Complementary Items; Components: Order Placement
FR-02.9	Mark complementary items as free of cost	Sequence Diagram: Place Order – markAsComplementary(); Classes: OrderItem (isComplementary attribute), ComplementaryItem; Components: Order Placement
FR-02.10	Generate and display order summary with all details	DFD Level 2.1: Process 2.6 “Generate Order Summary”; Sequence Diagram: Place Order – getOrderSummary(); Classes: Order, Bill; Components: Order Placement

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FR-02.11	Forward confirmed order to cashier and kitchen staff	DFD Level 2.1: Process 2.7 “Confirm & Submit Order”; DFD Level 2.2: Order flow to cashier and kitchen; Sequence Diagram: Send Order to Kitchen – sendOrderToKitchen(), forwardOrderToCashier(); Activity Diagram: Order Workflow – Step 8 “Forward Order to Cashier & Kitchen”; Classes: OrderManagementService, KitchenService, BillingService; Components: Order Processing
FR-03.1	Assign unique order ID to each confirmed order	Sequence Diagram: Send Order to Kitchen – assignUniqueOrderId(); Activity Diagram: Order Workflow – Step 7 “Generate Unique Order ID”; Classes: Order (orderId attribute); Components: Order Processing
FR-03.2	Transmit order details to cashier and kitchen	DFD Level 2.2: Process 2.7 showing order transmission; Sequence Diagram: Send Order to Kitchen – complete transmission flow; Activity Diagram: Order Workflow – parallel activities; Classes: OrderManagementService, KitchenService, BillingService; Components: Order Processing

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FR-03.3	Set initial status to “Order Pending”	DFD Level 2.2: Process 2.9 initial status; Sequence Diagram: Send Order to Kitchen – setStatus(ORDER_PENDING); Activity Diagram: Order Workflow – initial status setting; Classes: Order, OrderStatus enum (ORDER_PENDING); Database: D4 Order Status; Components: Order Processing
FR-03.4	Allow modifications or cancellations before kitchen acceptance	DFD Level 2.2: Process 2.9 “Process Order Cancellation”; Sequence Diagram: Modify or Cancel Order – status validation; Activity Diagram: Order Modification and Cancellation Workflow – pre-acceptance flow; Classes: OrderManagementService, Order; Components: Order Processing
FR-03.5	Prevent modification or cancellation after kitchen acceptance	DFD Level 2.2: Process 2.8 “Process Order Modification” with lock status; Sequence Diagram: Modify or Cancel Order – status check preventing changes; Activity Diagram: Order Modification and Cancellation Workflow – “Not Allowed” alternative flow; Classes: OrderManagementService, Order; Components: Order Processing

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FR-03.6	Allow additional items to be added after kitchen acceptance	Sequence Diagram: Modify or Cancel Order – loop for adding new items; Activity Diagram: Order Modification and Cancellation Workflow – “Add New Items” flow; Classes: OrderManagementService, Order; Components: Order Processing
FR-03.7	Track and update order status through lifecycle	DFD Level 2.2: Process 2.10 “Update Order Status”; Activity Diagram: Order Workflow – complete status transitions; Classes: Order, OrderStatus enum (ORDER_PENDING, IN_PREPARATION, PREPARED, SERVED, COMPLETED); Database: D4 Order Status; Components: Order Processing
FR-03.8	Display updated status to waiter, cashier, and kitchen	DFD Level 2.2: Status update flows; Sequence Diagram: Send Order to Kitchen – notifications; Classes: NotificationService; Components: Order Processing
FR-04.1	Display all pending orders to kitchen staff	Sequence Diagram: Send Order to Kitchen – display pending orders; Activity Diagram: Order Workflow – Step 9a “Kitchen Receives Order”; Classes: KitchenService, Kitchen, KitchenOrder; Components: Kitchen Operations

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FR-04.2	Allow kitchen staff to accept assigned order	Sequence Diagram: Send Order to Kitchen – acceptOrder(), addToPendingOrders(); Activity Diagram: Order Workflow – Step 10a “Kitchen Staff Accepts Order”; Classes: KitchenService, KitchenOrder; Components: Kitchen Operations
FR-04.3	Update status to “In Preparation” when accepted	Sequence Diagram: Send Order to Kitchen – updateStatus(IN_PREPARATION); Activity Diagram: Order Workflow – Step 11a “Status = In Preparation”; Classes: Order, OrderStatus (IN_PREPARATION), KitchenService; Components: Kitchen Operations
FR-04.4	Prevent modifications or cancellations once accepted	DFD Level 2.2: Process 2.8 lock status mechanism; Sequence Diagram: Modify or Cancel Order – status validation; Activity Diagram: Order Modification and Cancellation Workflow – “Not Allowed” flow; Classes: OrderManagementService; Components: Kitchen Operations
FR-04.5	Allow kitchen to mark order as “Prepared”	Activity Diagram: Order Workflow – Step 13a “Mark Order as Prepared”; Classes: KitchenService, Kitchen; Components: Kitchen Operations

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FR-04.6	Notify waiter when order marked as “Prepared”	Sequence Diagram: Send Order to Kitchen – notifyKitchenStaff(); Activity Diagram: Order Workflow – Step 14a “Notify Waiter Order Ready”; Classes: NotificationService; Components: Kitchen Operations
FR-05.1	Notify waiter when order is prepared	Sequence Diagram: Feedback submission showing notification flow; Activity Diagram: Order Workflow - Step 15 “Waiter Receives Notification”; Classes: NotificationService, Waiter; Components: Order Serving
FR-05.2	Allow waiter to mark order as “Served”	Activity Diagram: Order Workflow – Step 16 “Waiter Marks Order as Served”; Classes: OrderManagementService, Waiter; Components: Order Serving
FR-05.3	Update status to “Served” when confirmed	Activity Diagram: Order Workflow – Step 17 “Update Status = Served”; Classes: Order, OrderStatus (SERVED); Components: Order Serving
FR-05.4	Allow waiter to mark order as “Completed”	Activity Diagram: Order Workflow – Step 19 “Waiter Marks Order as Completed”; Classes: OrderManagementService, Waiter; Components: Order Serving
FR-05.5	Update status to “Completed” and display to dashboards	Activity Diagram: Order Workflow – Step 20 “Update Status = Completed” with note “Order ready for billing”; Classes: Order, OrderStatus (COMPLETED); Components: Order Serving

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FR-06.1	Generate bill with all details including discounts and complementary items	Activity Diagram: Billing & Payment Workflow – Steps 1–10; Activity Diagram: Order Workflow – Step 21 “Cashier Generates Bill”; Classes: BillingService, Bill (subtotal, discount, tax, serviceCharge, totalAmount); Components: Billing and Payment
FR-06.2	Allow cashier to select payment method (cash, card, online)	Activity Diagram: Billing & Payment Workflow – Step 12 “Display Payment Method Options”; Activity Diagram: Order Workflow – payment method decision point; Sequence Diagram: Payment processing method selection; Classes: PaymentGateway, PaymentMethod enum (CASH, CARD_SWIPE, ONLINE_BANK, DIGITAL_WALLET); Components: Billing and Payment
FR-06.3	Validate payments based on selected method	Activity Diagram: Billing & Payment Workflow – Steps 13a–23b showing validation flows; Sequence Diagram: Payment validation for each method; Classes: PaymentGateway, BillingService; Components: Billing and Payment

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FR-06.4	Process payments securely and log transactions	Activity Diagram: Billing & Payment Workflow – Steps 14–19 payment processing and logging; Activity Diagram: Order Workflow – Step 26 “Log Transaction Details”; Sequence Diagram: Transaction logging flow; Classes: PaymentGateway, Payment; Components: Billing and Payment
FR-06.5	Generate payment receipt with complete details	Activity Diagram: Billing & Payment Workflow – Step 22 “Generate Payment Receipt”; Activity Diagram: Order Workflow – Step 25 “Generate Payment Receipt”; Classes: PaymentReceipt (receiptId, paymentId, orderDetails, amounts), BillingService; Components: Billing and Payment
FR-06.6	Store receipts and transactions for accounting/auditing	Activity Diagram: Billing & Payment Workflow – Step 26 “Log Transaction Details”; Activity Diagram: Order Workflow – Step 32 “Archive Payment & Maintain Records”; Classes: Payment, PaymentReceipt; Components: Billing and Payment

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FR-06.7	Allow customer feedback submission after payment	Activity Diagram: Billing & Payment Workflow – Step 27 “Display Feedback Form (Rating 1–5)”; Sequence Diagram: Feedback submission sequence; Activity Diagram: Order Workflow – Step 27 “Display Feedback Form”; Classes: FeedbackService; Components: Customer Feedback
FR-07.1	Allow feedback submission after successful payment	Activity Diagram: Billing & Payment Workflow – Steps 27–29 feedback flow; Sequence Diagram: Complete feedback submission flow; Activity Diagram: Order Workflow – feedback steps; Classes: FeedbackService, Feedback; Components: Customer Feedback
FR-07.2	Provide rating scale 1–5	Activity Diagram: Billing & Payment Workflow – Step 27 “Rating 1–5”; Sequence Diagram: Rating scale interface; Classes: Feedback (rating: Integer); Components: Customer Feedback
FR-07.3	Store feedback with order information	Activity Diagram: Billing & Payment Workflow – Step 29c “Save Feedback with Order ID & Timestamp”; Sequence Diagram: storeFeedback() method; Classes: Feedback (orderId, rating, comments, timestamp); Components: Customer Feedback
FR-08.1	Generate detailed sales reports (daily, weekly, monthly)	Classes: ReportingService, SalesReport; Components: Administration

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FR-08.2	Include order count, sales, discounts, complementary values, payment summaries	Classes: SalesReport (totalOrders, totalSales, totalDiscounts, complementaryValue, paymentBreakdown); Components: Administration
FR-08.3	Identify most-selling and low-performing items	Classes: ReportingService, MenuManagementService; Components: Administration
FR-08.4	Allow admin to add/update/remove menu items	Classes: MenuManagementService, Admin; Components: Administration
FR-08.5	Allow admin to manage complementary items list	Classes: MenuManagementService, ComplementaryItem; Components: Administration
FR-08.6	Allow admin to manage user accounts	Classes: UserManagementService (addUser, updateUser, deactivateUser, removeUser), Admin; Components: Administration
FR-08.7	Display error messages for invalid input/operations	Activity Diagram: Error handling flows in all workflows; Classes: All service classes with error handling; Components: All Modules